Block practice · Embedded systems, cloud storage, software and translators

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Embedded systems: small computers inside devices

Cloud storage: what it is, pros, cons and security

What the operating system does, and system vs application software

High-level and low-level languages

Translators: assembler, compiler and interpreter

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Practice workbook · Embedded systems: small computers inside devices

What an embedded system is, its characteristics, examples inside everyday devices, the sensors and other hardware it controls, and why it suits a dedicated task.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-09

1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a.ii only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

DO NOT WRITE IN THIS AREA

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

.....

2 .....

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(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

1. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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2. Question 6 · part a.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

.....

.....

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P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

6 A train company uses ticket vending machines at each station.

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1 .....

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.....

.....

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P 6 1 8 8 0 A 0 2 0 2 4

3. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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4. Question 5 · part c.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.i only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

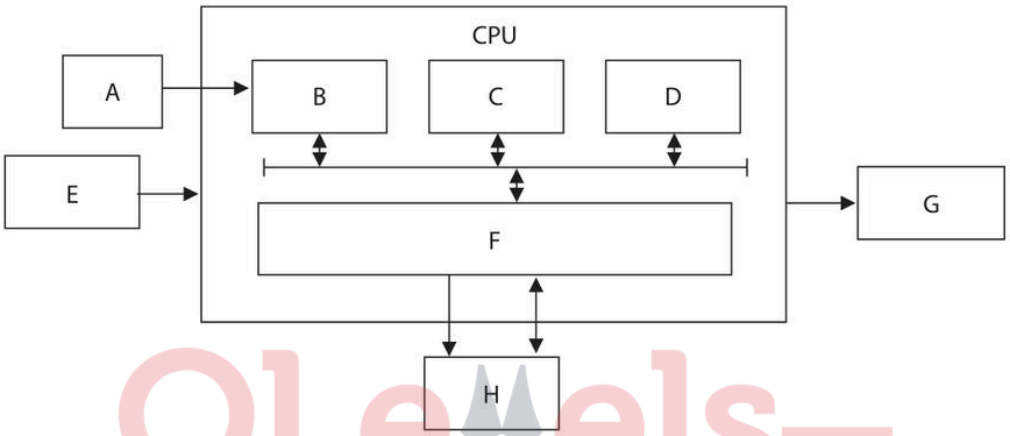


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 5 · part c.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.i only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

☒ **A** Memory delete signal

☒ **B** Memory read signal

☒ **C** Memory store signal

☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1 .....

2 .....

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4. Question 5 · part c.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.i only.

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(ii) There are many factors to consider when choosing a CPU for an embedded system.

For example, the clock speed of the CPU in a TV remote control can be low because it only has to be fast enough to create a low frequency output signal for the TV.

Explain **one other** factor that should be considered when choosing a CPU for an embedded system within a TV remote control.

(2)

(Total for Question 5 = 13 marks)

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Working and answer

5. Question 5 · part c.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

5 Computers can be general-purpose or specific-purpose machines.

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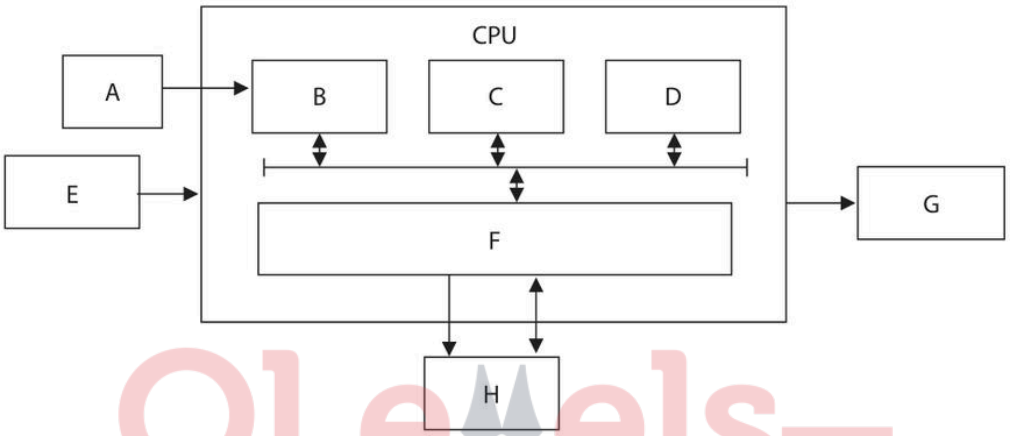


Figure 4

Component	Letter
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Output device	
Registers	C
Main memory	
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Input device	
Arithmetic/logic unit (ALU)	
Clock	A

DO NOT WRITE IN THIS AREA
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5. Question 5 · part c.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

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One register is the program counter. Its function is to keep track of the next instruction to be fetched.

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State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

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(2)

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5. Question 5 · part c.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

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For example, the clock speed of the CPU in a TV remote control can be low because it only has to be fast enough to create a low frequency output signal for the TV.

Explain **one other** factor that should be considered when choosing a CPU for an embedded system within a TV remote control.

(2)

(Total for Question 5 = 13 marks)

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Working and answer

6. Question 5 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

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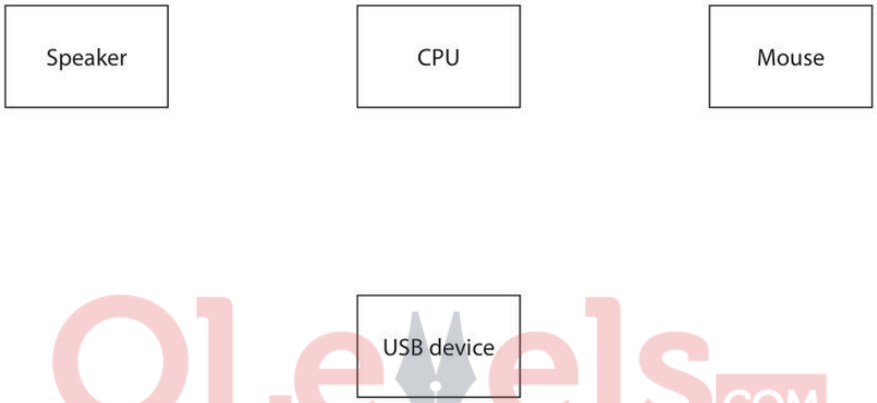
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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6. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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P 7 2 9 3 7 A 0 1 7 2 4

17

Turn over ▶

6. Question 5 · part c · 2 marks · Working and answer

Working and answer

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7. Question 5 · part a.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.i only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

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2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

P 7 8 6 3 7 A 0 1 3 2 0

13

Turn over ▶

7. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Cloud storage: what it is, pros, cons and security

What cloud storage is, its benefits and drawbacks, and its security issues including how protection duties are split between user and provider, from 1-mark items to a 6-mark discussion.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-08

1. Question 2 · part b.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

.....

2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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1. Question 2 · part b.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

.....

2 .....

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3 .....

.....

(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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2 .....

.....

(Total for Question 2 = 12 marks)

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1. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

.....

.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	**Zafer**
Washing machine design drawings		
A folder of manuals for new machines		

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2. Question 2 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b.ii · 2 marks

2022 JUN · Paper 01

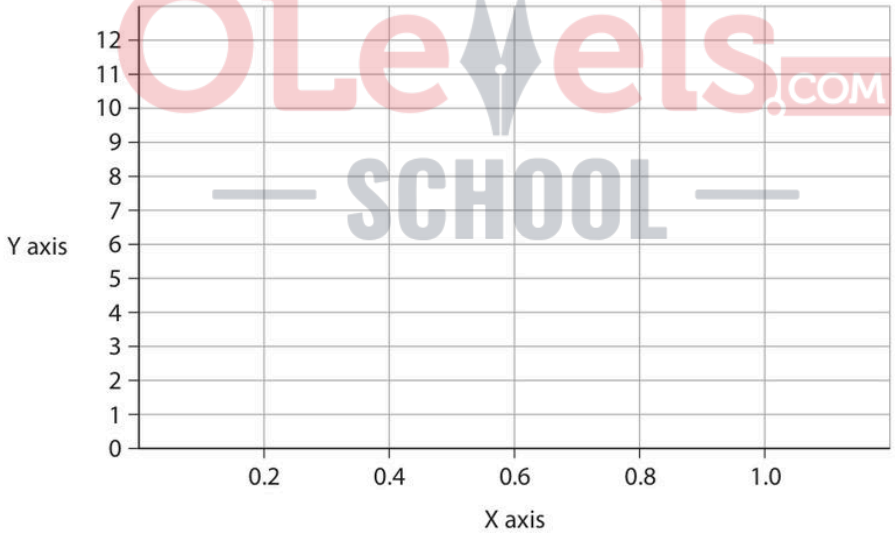
Further practice · Attempt part b.ii only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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P 7 2 5 3 8 A 0 8 2 0

3. Question 3 · part b.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

.....

.....

.....

.....

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(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

.....

.....

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

3. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 3 · part b.iii · 1 mark

2022 JUN · Paper 01

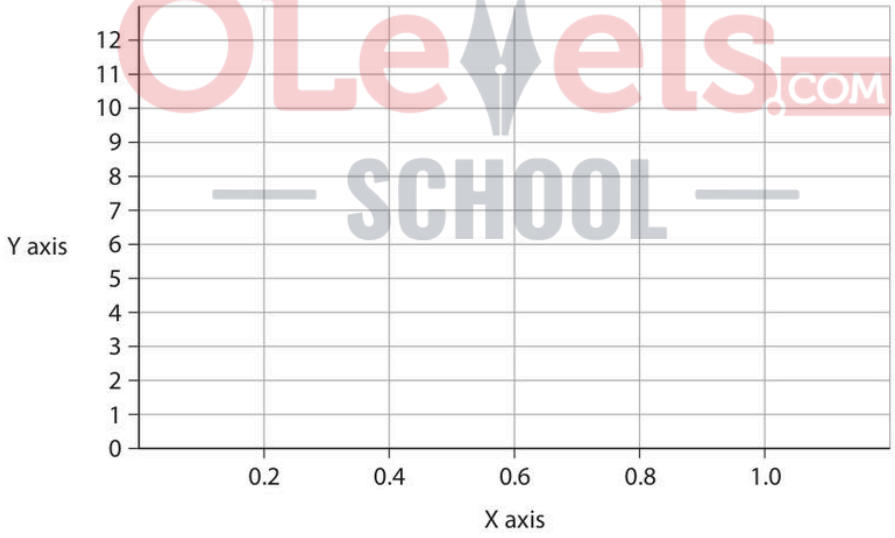
Further practice · Attempt part b.iii only.

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3	12
4	5
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Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

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P 7 2 5 3 8 A 0 8 2 0

4. Question 3 · part b.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(b) Alyssa uploads music files to her cloud storage.

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Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

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.....

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.....

.....

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Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

4. Question 3 · part b.iii · 1 mark · Working and answer

Working and answer

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5. Question 6 · part d · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 6 · part d · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

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(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

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(ii) State the purpose of the algorithm in **Figure 2**.

(1)

5. Question 6 · part d · 6 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

(d) Cloud storage has become an increasingly popular service.

Discuss the benefits and drawbacks of using cloud storage.

(6)

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(Total for Question 6 = 12 marks)

5. Question 6 · part d · 6 marks · Working and answer

Working and answer

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Practice workbook · What the operating system does, and system vs application software

What the operating system does (process, memory, file, peripheral and user-interface management, multitasking, file storage on disc) and the difference between system and application software.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

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Type ID: CSC-T4-12

1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

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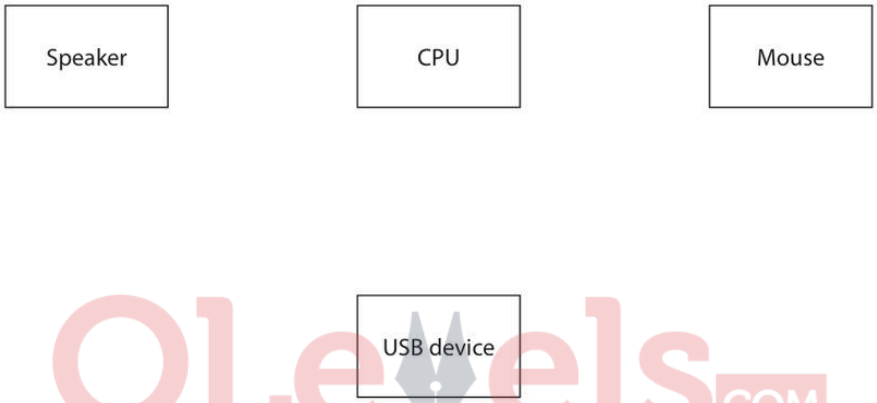
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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P 7 2 9 3 7 A 0 1 7 2 4

17

Turn over ▶

1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

(ii) A warehouse has an automated alarm system.

When the alarm system is activated it will sound if:

- a movement sensor (M) is activated
- a pressure pad (P) is activated
- a key code (C) to deactivate the alarm system has not been entered.

Construct a logic statement, using AND, OR and NOT with the letters M, P and C, to show the conditions that will sound the alarm.

(3)

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1. Question 5 · part e · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part e only.

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(e) A computer with a single CPU runs several processes at the same time.

This computer is multitasking.

Describe how the operating system enables processes to share a single CPU.

(2)

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.....

(Total for Question 5 = 18 marks)

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Working and answer

2. Question 1 · part f · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part f only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

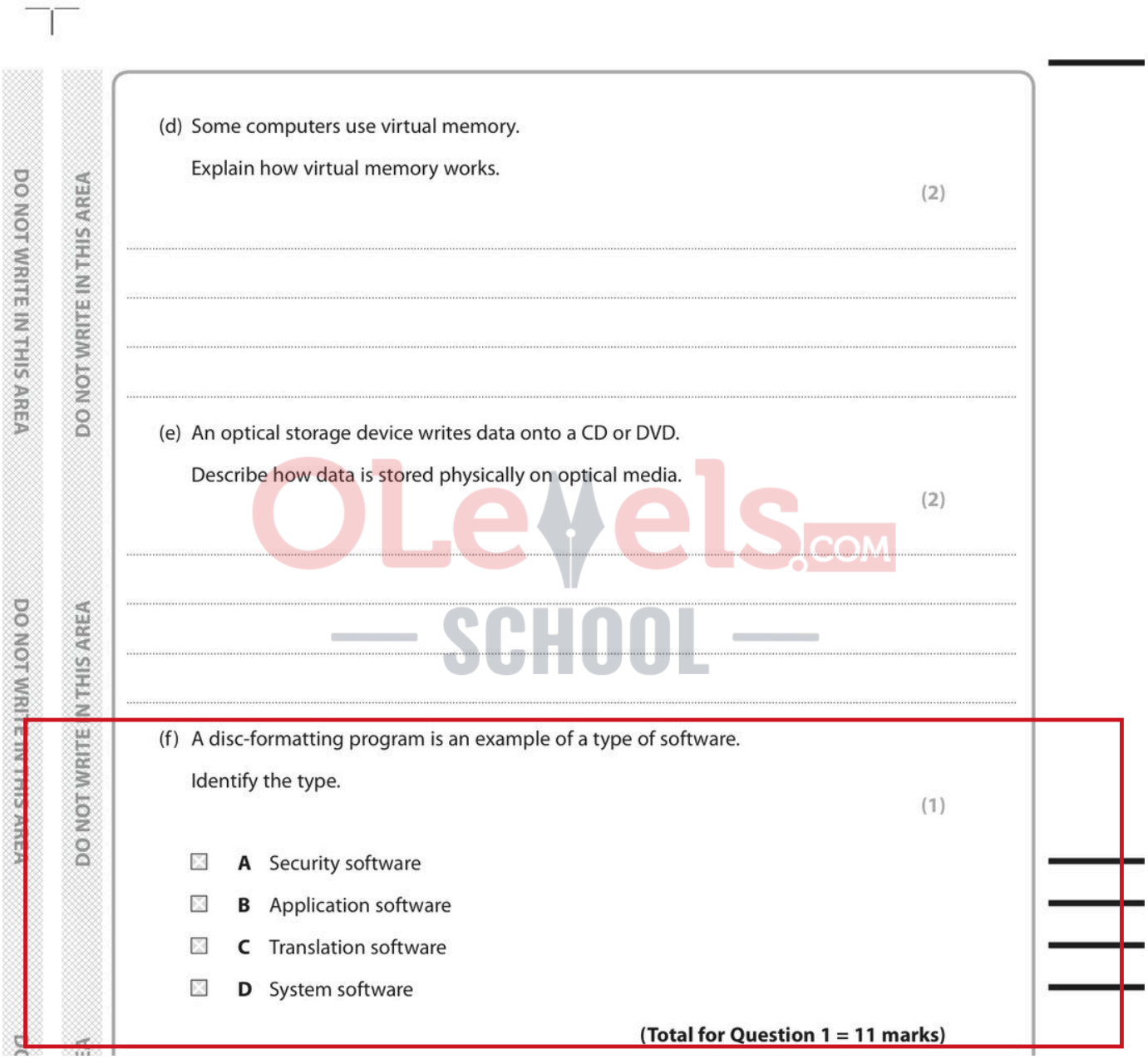
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2. Question 1 · part f · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part f only.



(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

(Total for Question 1 = 11 marks)

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2. Question 1 · part f · 1 mark · Working and answer

Working and answer

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3. Question 3 · part d · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

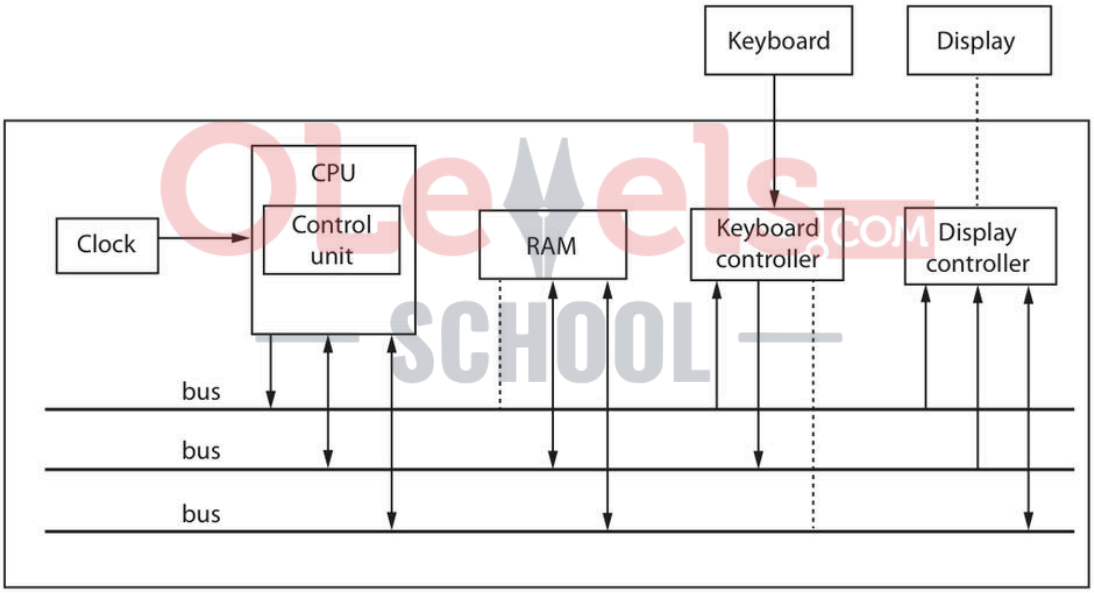
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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3. Question 3 · part d · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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(c) Some computers have more than one CPU. The CPUs work in parallel to execute instructions.

Complete the table to show how two CPUs working in parallel could execute the expression:

(5 * 2) + (4 * 3) – (24 / 4)

(4)

CPU 1	CPU 2
5 * 2 = 10	

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(d) One function of an operating system is to manage files on secondary storage.

Describe how an operating system manages the storage of a file on random access secondary storage.

(4)

(Total for Question 3 = 16 marks)

3. Question 3 · part d · 4 marks · Working and answer

Working and answer

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4. Question 3 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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P 7 2 9 3 7 A 0 8 2 4

4. Question 3 · part c · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

.....

.....

.....

.....

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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4. Question 3 · part c · 2 marks · Working and answer

Working and answer

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5. Question 2 · part a.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a.i only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

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5. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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6. Question 2 · part a.ii · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a.ii only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

1 .....

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2 .....

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3 .....

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6. Question 2 · part a.ii · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a.ii only.

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(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

(2)

(ii) Describe how data is read from a magnetic hard drive.

(3)

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(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

Stage	Sequence number
The instruction is placed onto the data bus	
The Arithmetic and Logic Unit (ALU) performs a calculation	
The address of the next instruction is placed onto the address bus	
The instruction is decoded by the control unit	

(4)

6. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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Practice workbook · High-level and low-level languages

Characteristics of high- and low-level languages, their benefits and drawbacks, and why a task suits one or the other.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-15

1. Question 5 · part b.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

1. Question 5 · part b.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

1. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 6 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

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P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(b) Compare **four** features of high-level and low-level programming languages.

(4)

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2. Question 6 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Artificial intelligence (AI), in many forms, has an increasing impact on our lives.

Discuss this statement considering characteristics, uses and ethical issues.

(6)

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(Total for Question 6 = 15 marks)

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2. Question 6 · part b · 4 marks · Working and answer

Working and answer

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3. Question 6 · part a.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a.i only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

.....

.....

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

.....

.....

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

.....

2 .....

.....

(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

.....

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3. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 6 · part a.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a.ii only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

.....

.....

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

.....

.....

(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

.....

2 .....

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(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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4. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.i only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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5. Question 1 · part e.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.i only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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5. Question 1 · part e.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Translators: assembler, compiler and interpreter

What each translator does, how compilers and interpreters differ, their advantages, and choosing a translator for a task.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-16

1. Question 5 · part a · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator		
	Assembler	**Interpreter**	**Compiler**
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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P 7 5 7 3 6 A 0 1 8 2 8

1. Question 5 · part a · 3 marks · Working and answer

Working and answer

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2. Question 6 · part c · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

6 A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

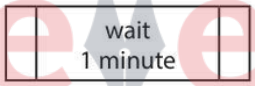
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.



(6)

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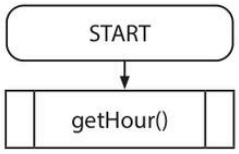
2. Question 6 · part c · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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Draw your flowchart here.



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2. Question 6 · part c · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

Figure 4 shows this algorithm written in pseudocode.

```
 1 SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
 2
 3 SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
 4
 5 SET i TO 0
 6
 7 WHILE i < 8 DO
 8
 9     IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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Figure 4

(i) Give the output produced by the algorithm.

(2)

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

(1)

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2. Question 6 · part c · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

(iii) State how the pseudocode needs to be changed to make this amendment.

(1)

.....

.....

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QUESTION 6(c) BEGINS ON THE NEXT PAGE.

2. Question 6 · part c · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

(c) The algorithms will be coded in a high-level language.

Programs written in high-level languages must be translated in order to be executed by a computer.

Discuss the similarities and differences between a compiler and an interpreter.

(6)

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(Total for Question 6 = 16 marks)

2. Question 6 · part c · 6 marks · Working and answer

Working and answer

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3. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

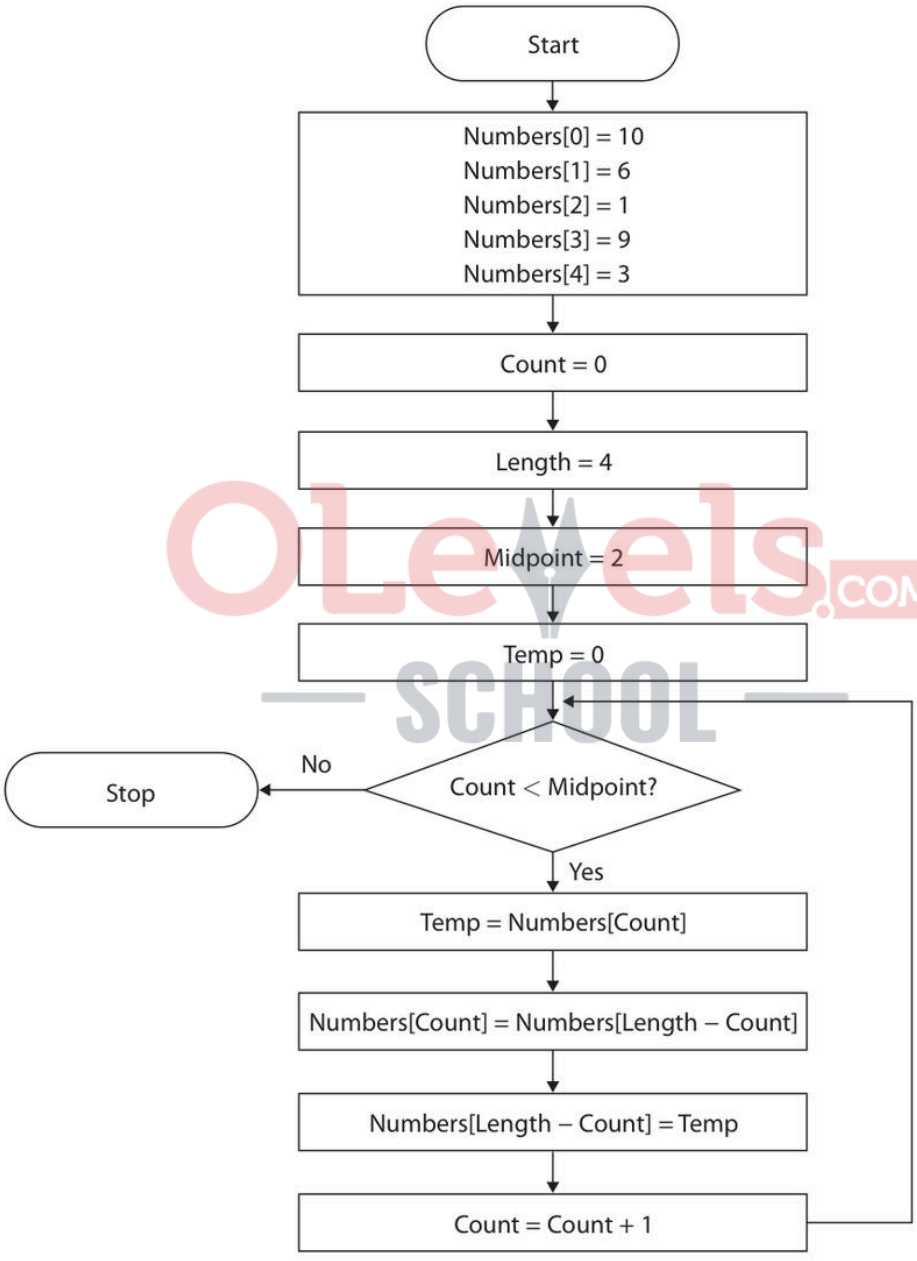


Figure 3

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P 6 6 4 8 9 A 0 1 2 2 0

3. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

.....

.....

(iii) Explain why the variable Temp is needed.

(2)

.....

.....

.....

.....

3. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1    SET total TO 0
2    SET number TO 0
3    SET count TO 0
4    WHILE number <> –1 DO
5       SEND 'Input a number or –1 to end the program' TO DISPLAY
6       RECEIVE number FROM (INTEGER) KEYBOARD
7       SET total TO total + number
8       SET count TO count + 1
9    END WHILE
10   SET average TO total / count
11   SEND 'The average is ' & average TO DISPLAY
```

Figure 4

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
The average is 2.75	The average is 2.2

(i) Explain why the **Actual** result is not the same as the **Expected** result.

(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

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P 6 6 4 8 9 A 0 1 4 2 0

3. Question 5 · part c · 2 marks

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Further practice · Attempt part c only.

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(c) Isaac is going to sell one of his programs and distribute it on a DVD.

Give **two** advantages of choosing a programming language that uses a compiler, rather than an interpreter, for a program that will be distributed on a DVD.

(2)

1 .....

.....

2 .....

.....

(Total for Question 5 = 14 marks)

Working and answer

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4. Question 1 · part e.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.ii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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4. Question 1 · part e.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.ii only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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4. Question 1 · part e.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.iii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part e.iii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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5. Question 1 · part e.iii · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part e.iii only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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5. Question 1 · part e.iii · 3 marks · Working and answer

Working and answer

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6. Question 3 · part f · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part f only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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6. Question 3 · part f · 4 marks

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Further practice · Attempt part f only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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6. Question 3 · part f · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part f only.

(e) Here is part of a file that contains electric meter readings.

04631 04984 05103 05163 05271 05383 05487 05722

Explain the effect on the file of applying a run-length encoding algorithm to this data file.

(2)

.....

.....

.....

.....

(f) A series of tasks is written in different programming languages.

Complete the table to show the correct programming language translator for each task.

(4)

Task	Translator
A guessing game that can be used on different computing platforms	
A screen driver for a new smartphone	
A new version of a spreadsheet program for sale next year	
Control software for an embedded system inside a new washing machine	

(Total for Question 3 = 15 marks)

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6. Question 3 · part f · 4 marks · Working and answer

Working and answer

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7. Question 5 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

1 .....

.....

2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

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7. Question 5 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

7. Question 5 · part b.i · 1 mark · Working and answer

Working and answer

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